

PROFESSIONAL SUMMARY

Senior Data Scientist with 5+ years of experience building GenAI, ML, and real-time data systems across fintech, healthcare, and IoT domains. Expertise in LLMs (GPT, LLaMA), RAG, LangChain/LangGraph, and vector databases (FAISS, Pinecone) for fraud detection and risk intelligence. Strong in end-to-end ML pipelines and large-scale data engineering using Python, SQL, PySpark, Spark Streaming, Kafka, and Databricks, with hands-on experience in predictive modeling, anomaly detection, MLOps, and cloud deployment (AWS, Docker, Kubernetes) delivering scalable, production-grade AI systems.

SKILLS

Machine Learning & Statistics: Supervised/Unsupervised Learning, Regression, Classification, Clustering, XGBoost, LightGBM, Random Forest, Time Series Forecasting, Feature Engineering, Model Evaluation (ROC-AUC, Precision-Recall, F1-score), A/B Testing, Anomaly Detection

AI / GenAI & Deep Learning: LLMs (GPT, LLaMA), LangChain, LangGraph, RAG Architecture, Prompt Engineering, Agentic AI, Transformers, Hugging Face, OpenAI API, Embeddings, Vector Databases (Pinecone, FAISS), Explainable AI (SHAP)

Big Data & Cloud: Apache Spark, PySpark, Spark Streaming, Kafka, Databricks, AWS (S3, Lambda), Real-time Data Processing, Distributed Systems

MLOps & Deployment: Docker, Kubernetes, CI/CD, MLflow, Airflow, Model Monitoring, Automated Retraining, Model Serving (Batch & Real-time)

Data Engineering & SQL: Advanced SQL (Query Optimization), ETL Pipelines, Data Modeling, PySpark Data Pipelines, Data Quality Checks

Visualization & BI: Power BI (DAX), Tableau, Looker, KPI Dashboards, Executive Reporting

WORK EXPERIENCE

JP Morgan Chase & Co.

Senior Data Scientist

Dallas, Texas

Jul2024 – Present

- Built a GenAI Financial Risk Intelligence Copilot using GPT/LLaMA with RAG architecture, enabling analysts to query 300K+ fraud transaction records in natural language and reducing manual investigation effort across enterprise risk review workflows.
- Designed a vector-based knowledge retrieval system using FAISS/Pinecone and embedding models, improving contextual accuracy of fraud and credit risk explanations by 40% compared to traditional rule-based lookup systems.
- Developed real-time data pipelines using Kafka, Spark Streaming, and Snowflake, processing 5M+ financial events daily and accelerating fraud risk detection through near real-time transaction monitoring and risk signal generation.
- Implemented LangChain/LangGraph-based LLM orchestration pipelines with RAG-driven transaction retrieval, enabling multi-step reasoning for fraud explanation, merchant profiling, and anomaly detection with structured, explainable outputs for real-time risk analysis systems.
- Engineered feature enrichment layer combining transactional history, behavioral signals, and merchant risk attributes using Python (Pandas, NumPy) and SQL, improving risk context completeness by 30%.
- Established AI governance and explainability frameworks using SHAP-based model interpretation and audit-ready validation workflows, improving transparency and regulatory compliance for risk analytics models.
- Deployed containerized ML and LLM services on AWS (S3, Lambda, ECS) with CI/CD pipelines, MLflow model tracking, automated model monitoring, drift detection, and API-based serving, supporting scalable low-latency inference across enterprise fraud investigation platforms.

CVS Health

Data Scientist

Irving, Texas

Sep2023 – May 2024

- Architected an end-to-end patient risk scoring system for chronic disease prediction (diabetes, COPD, heart disease, hypertension) using Python, SQL, and PySpark, processing 2M+ EHR and claims records to identify high-risk patients via scalable healthcare data pipelines.
- Developed predictive classification models (XGBoost, LightGBM, Logistic Regression baseline) for chronic disease risk scoring, improving early identification of high-risk patients across multi-condition cohorts in large-scale healthcare datasets.
- Implemented survival analysis models using Kaplan-Meier estimators and Cox Proportional Hazards techniques, enabling time-to-event prediction and patient risk stratification across 90K+ longitudinal patient records, supporting data-driven clinical outcome analysis and treatment planning.
- Engineered advanced feature pipelines using Python (Pandas, NumPy), SQL, PySpark, and Databricks on EHR data including diagnosis history, lab results, prescriptions, and comorbidities, generating 500+ clinical risk features for large-scale predictive modeling and patient risk stratification.
- Integrated model explainability using SHAP values to provide interpretable risk drivers for clinicians, improving transparency in patient-level predictions and enabling doctor-facing decision support insights for high-risk cases.
- Deployed production ML pipelines using Airflow orchestration, SQL-based data workflows, and Spark batch processing, enabling automated model retraining and daily patient risk scoring at scale in healthcare environments.

KPIT

Data Scientist

India

Jun2020 – Jul 2022

- Developed an end-to-end predictive maintenance system for EV & ICE fleets using Python, SQL, PySpark, and PyTorch, processing 3M+ IoT telemetry records (engine temperature, vibration, battery health, sensor logs) to predict early failure patterns across fleet systems.
- Formulated advanced feature pipelines (500+ engineered signals) using time-window aggregations, rolling statistics, lag features, and missing value imputation on raw IoT sensor streams, improving model signal quality for downstream ML training.
- Developed and tuned machine learning models using Scikit-learn (Random Forest, Gradient Boosting, K-Means clustering) to detect anomalous vehicle behavior and classify failure risk levels across multi-component systems.
- Implemented time-series anomaly detection and forecasting models on critical sensor metrics (vibration, temperature, battery degradation), enabling early fault detection up to 7–14 days in advance of failure events.
- Evaluated model performance using AUC-ROC, Precision-Recall, F1-score, and confusion matrix analysis, optimizing thresholds for imbalanced failure datasets to reduce false negatives in high-risk vehicle predictions.
- Deployed scalable ML inference pipelines using AWS (S3, EC2), Spark batch processing, and Kubernetes-based containerized workflows, enabling 30K+ fleet-level daily predictions with automated retraining and scheduled scoring pipelines.

EDUCATION

Master of Science in Computer and Information Science | University of North Texas, Denton, TX

May 2024